

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

8-layer 20×20mm FPD-Link III automotive camera module PCB (image sensor + serializer + power)

Populated PCB (PCBA) · from photo analysis · Built in: China (Shenzhen / Suzhou) · Delivered: UK, duty paid · Currency: GBP

Delivered cost / board	Components	Bare board	Assembly & test	Freight & duty
£15.26	87.4%	2.7%	5.4%	3.5%

Estimate Confidence: **Low** · likely £12.22 – £19.77 · 11 BOM lines, 1 to verify with a quote

Key Assumptions

Populated board built in China (Shenzhen / Suzhou), delivered to the UK, import duty paid, at 250,000 boards a year.

Board: Automotive camera board · costed at ASIL-B · parts list from the BOM image (11 lines, 80 parts).

Component prices follow the parts bought (qty per board × boards); fabricator and EMS prices carry their own overhead and margin — none is added on top.

£1 — Cost Breakdown per Board

China (Shenzhen / Suzhou) · delivered UK, duty paid

Element	Amount	% of total	Basis
Components (as the EMS buys them)	£12.70	83.2%	11 BOM lines at China (Shenzhen / Suzhou) sourcing — see §3
EMS material burden (5%)	£0.64	4.2%	Procurement, inventory, scrap and margin on material — stated rate by volume
Bare PCB fabrication	£0.27	1.8%	Fabricator price: base, layers, finish, vias, set-up — see §4
Automotive fab grade	£0.14	0.9%	IATF 16949 line, automotive laminate, class 3 inspection, coupons
SMT / THT assembly	£0.26	1.7%	Placements and joints at the country line rate
Test & inspection	£0.39	2.6%	AOI, X-ray, ICT (station time + fixture over the order)
Automotive assembly grade	£0.18	1.2%	IATF line, class 3 workmanship, serialisation
Energy, packaging, cost of quality	£0.13	0.9%	Energy £0.01 · ESD packaging £0.06 · rework/scrap £0.06
Freight to the UK	£0.00	0.0%	Sea at volume, air below 2,500 boards

Element	Amount	% of total	Basis
UK import duty	£0.54	3.5%	On components + board + assembly (customs value)
Rounding	£0.01	0.1%	Each figure is rounded to the penny on the server
Delivered cost per board	£15.26	100.0%	China (Shenzhen / Suzhou) build, delivered UK, duty paid

Components, bare board and assembly are supplier prices: the fabricator's and the EMS's overhead and margin are inside them, so no further overhead or margin is added.

§2 — Basis & Confidence

Build country	China (Shenzhen / Suzhou)	Annual volume	250,000 boards
Delivery	UK, import duty paid	Board type	Automotive camera board
Parts list from	BOM image	Lines · parts	11 lines · 80 parts
Estimate confidence	Low	Likely range	£12.22 – £19.77
Lines to verify	1 (£4.64)	Meaning	£1.00+ lines with no distributor price

Functional Safety (ISO 26262)

Costed as ASIL-B (the photo classifier said ASIL-C); quality grade auto grade 2. An ASIL comes from the hazard analysis (HARA), not from a photograph — confirm it against the safety concept.

The classifier described a radar module, but the parts list is a camera board — its description is withheld as inconsistent with the BOM.

ASIL-C was claimed from the photos, but the parts list has no safety PMIC / SBC or lockstep safety MCU that an ASIL-C design carries — costed at ASIL-B (no burn-in). Confirm the level against the safety concept; an ASIL-C answer adds burn-in and the higher NRE tier.

§3 — Bill of Materials11 lines · 80 parts · £12.70

#	Description / part number	Package	Qty	Unit	Ext	Priced from
1	Ceramic Capacitor	<= 1206	49	£0.02	£0.91	class range
2	Ferrite		7	£0.04	£0.30	class range
3	Common Mode Choke		1	£0.24	£0.24	class range
4	Common Mode Choke		1	£0.24	£0.24	class range
5	FPD-Link III Serializer With CSI-2 Interface DS90UB935-Q1	VQFN-32	1	£1.87	£1.87	catalogue (distributor)
6	Automotive Step Down DC-DC Converter LM53600MQDSXRQ1	WSOP-10	1	£1.31	£1.31	catalogue (distributor)
7	2.25-MHz Fixed VOUT Dual 1000mA/400mA Step-Down TPS62422QDRCRQ1	MSOP-10	1	£0.82	£0.82	catalogue (distributor)
8	300-mA, Low-IQ, Low-Dropout Regulator TLV70233QDSERQ1	WSOP-6	1	£0.17	£0.17	catalogue (distributor)
9	LI-ION AND LI-POL CHARGER IC BQ24025DRCR	VSON(DRC)10	1	£2.10	£2.10	catalogue (distributor)
10	Resistor	<= 1206	16	£0.01	£0.08	class range

#	Description / part number	Package	Qty	Unit	Ext	Priced from
11	Image sensor		1	£4.64	£4.64	class range *
COMPONENTS TOTAL			80		£12.70	

The 11 lines (80 parts) total £12.70 — the components row of \$1. Unit prices are the distributor price × 0.879, China (Shenzhen / Suzhou)'s component sourcing index. The EMS material burden (£0.64) is added on top in \$1.

Source: catalogue (distributor) = a distributor price on file; catalogue (estimate) = a dated engineering estimate; named-part range = the part is identified, priced inside its own range; function / class range = priced from the tool's table for that kind of part. Lines marked * are £1.00+ with no distributor price behind them — confirm with a quote.

§4 — Board & Assembly

Size	20 × 20 mm (estimated from the photos)
Layers	8
Technology / HDI	FR4 HTg
Finish · copper	OSP · 2 oz
Vias (through / blind / micro)	115 / 0 / 0
Impedance control	Yes
Placements · BGA/CSP	80 SMT · 1 BGA/CSP · 2 reflow side(s)
Panel	280 boards on 480 × 350 mm (67% used)
Bare-board build-up	base £0.01 · layers £0.03 · vias £0.22 · copper £0.01 · impedance £0.01

§5 — Build-Country Comparison

each row: this board built there, delivered UK, duty paid

Country	Components	Bare board	Assembly	Other	Freight + duty	Total	vs China	Lead
China	£13.34	£0.41	£0.83	£0.14	£0.54	£15.26	costed	3 wk
Vietnam	£13.95	£0.49	£0.52	£0.13	£0.83	£15.92	+£0.66	3 wk
Malaysia	£13.80	£0.57	£0.92	£0.13	£0.84	£16.26	+£1.00	3 wk
Thailand	£14.10	£0.58	£0.79	£0.15	£0.85	£16.47	+£1.21	3 wk
India	£14.40	£0.52	£0.73	£0.17	£0.86	£16.68	+£1.42	4 wk
Taiwan	£13.65	£0.93	£1.76	£0.17	£0.61	£17.12	+£1.86	3 wk
Mexico	£14.86	£0.80	£1.16	£0.16	£0.63	£17.61	+£2.35	3 wk
South Korea	£13.95	£1.24	£2.72	£0.21	£0.00	£18.12	+£2.86	3 wk
Czech Republic	£15.92	£1.04	£1.62	£0.22	£0.00	£18.80	+£3.54	2 wk
Poland	£16.07	£0.94	£1.58	£0.21	£0.00	£18.80	+£3.54	2 wk
Germany	£17.89	£2.14	£4.17	£0.33	£0.00	£24.53	+£9.27	2 wk
USA	£17.44	£2.02	£3.95	£0.33	£0.86	£24.60	+£9.34	2 wk
United Kingdom	£18.50	£2.49	£3.75	£0.37	£0.00	£25.11	+£9.85	1.5 wk
Japan	£18.19	£3.53	£3.60	£0.35	£0.00	£25.67	+£10.41	3 wk

"Other" is energy, ESD packaging and cost of quality (rework and scrap at the country's defect rate). Components follow each country's sourcing index; labour, energy and duty are each country's own.

§6 — One-Time Automotive NRE

£16,200.00 · not in the unit cost

Item	One-time	Per board over one year (250,000)
PPAP	£2,600.00	£0.01
FMEA	£4,400.00	£0.02
DV / PV (DVP&R)	£7,000.00	£0.03
ASIL audit	£2,200.00	£0.01
Total NRE	£16,200.00	£0.06

§7 — Cost Drivers & Next Steps

What drives this board's cost

3 lines carry 56% of the board: Image sensor £4.64; LI-ION AND LI-POL CHARGER IC (BQ24025DR) £2.10; FPD-Link III Serializer With CSI-2 Interface (DS90UB935-Q1) £1.87. These are the quotes to get first. Components are 87% of the board; bare board 3%, assembly and test 5%. A should-cost discussion on this board is a component-price discussion. China (Shenzhen / Suzhou) is the lowest delivered cost of the 14 countries compared. 1 line of £1.00+ (£4.64) is priced from a range, not a quote: Image sensor.

Not in this unit cost

One-time automotive NRE (£16,200.00 — PPAP, FMEA, DV/PV, audit) is not in the unit cost; see §6. Stencils, test fixtures and programming are inside the EMS price, spread over the order (no separate tooling line). Tier-1 / distributor margin above the EMS price, and the enclosure, lens and cable of a module, are not in this board cost.

Analysis checks (1)

BOM taken from your BOM image (transcribed by the reader, never priced by it): 11 lines, 80 SMT placements. The photos were used for the board build and to fill gaps, not to write the BOM.

Analysis limitations

The image sensor model is unknown

Source Photographs

2 images the costing was built from

The parts list came from the BOM image supplied with the analysis; the photographs gave the board's size, layer and assembly reading and any package the list left open.